Lecture 10
Cluster Analysis

10.1

Contents

See Johnson and Wichern 2007, Applied Multivariate Statistical Analysis, Pearson Education.

10.2

1 Introduction

Goal
Find possible groupings of subjects (or variables) such that subjects (or variables) within groups are as more similar as possible and subjects (or variables) belonging to different groups are as different as possible.

Cluster analysis is an *exploratory method*.

The closeness measure between two objects, y_r and y_s (belonging to the p-dimensional space) is said a proximity measure. We can have:

- dissimilarity measures,
- similarity measures.

10.3

2 Dissimilarity and Similarity Measures

Dissimilarity Measure
Let $\mathbf{y}_r$ and $\mathbf{y}_s$ be two objects in the p-dimensional space.
A dissimilarity measure satisfies the following properties:

1. $d_{rs} \geq 0$
2. $d_{rs} = 0$ if $\mathbf{y}_r = \mathbf{y}_s$

3. $d_{rs} = d_{sr}$ (symmetric property)
4. The dissimilarity measure is said to be a *metric* when, in presence of a generic object $\mathbf{y}_q$ in the p-dimensional space, it holds

$$d_{rs} \leq d_{rq} + d_{qs} \text{ (triangular inequality).}$$

A dissimilarity measure decreases when objects are closer. 10.4

Examples of Dissimilarity Measures

- The *square* of the Euclidean distance

$$d_{rs}^2 = \|\mathbf{y}_r - \mathbf{y}_s\|^2 .$$

- The *square* of the weighted Euclidean distance

$$d_{rs}^2 = (\mathbf{y}_r - \mathbf{y}_s)' \left[\text{diag}(\boldsymbol{\Sigma})\right]^{-1} (\mathbf{y}_r - \mathbf{y}_s).$$

- The *square* of the Mahalanobis distance

$$d_{rs}^2 = (\mathbf{y}_r - \mathbf{y}_s)' \boldsymbol{\Sigma}^{-1} (\mathbf{y}_r - \mathbf{y}_s).$$

- The Minkowski distance

$$_M d_{rs} = \left(\sum_{j=1}^{p} \left| y_{rj} - y_{sj} \right|^{\lambda} \right)^{\frac{1}{\lambda}}$$

  - if $\lambda = 1$ we have the ℓ_1 (city block or Manhattan) distance.
  - if $\lambda = 2$ we have the ℓ_2 Euclidean distance.
  - if $\lambda \to \infty$ we have the Lagrange distance

$$d_{rs} = \max_j \left| y_{rj} - y_{sj} \right| .$$

10.5

Other particular distances exist.
We remind the Canberra distance

$$d_{rs} = \sum_{j=1}^{p} \frac{\left| y_{rj} - y_{sj} \right|}{(y_{rj} + y_{sj})}$$

that is used with asymmetric distributions or in presence of anomalous data. 10.6

Similarity Measure
Let $\mathbf{y}_r$ and $\mathbf{y}_s$ be two objects in the p-dimensional space.
A similarity measure satisfies the following properties:

1. $0 \leq s_{rs} \leq 1$
2. $s_{rs} = 1$ if $\mathbf{y}_r = \mathbf{y}_s$
3. $s_{rs} = s_{sr}$ (symmetric property)

A similarity measure increases when objects are closer.
An example of similarity measure is the Pearson's correlation distance. 10.7

3 Agglomerative Algorithms in Cluster Analysis

Two kind of aggregation methods are available

- Hierarchical Agglomerative
- Non-Hierarchical Agglomerative

10.8

Hierarchical Agglomerative
This is a 'bottom up' approach.

- Each observation starts in its own cluster.
- The most similar objects are first grouped, and these initial groups are merged according to their similarities. Eventually, as similarity decreases all subgroups are merged into a single cluster.
- If a subject is assigned to a group according to a certain criterion, it *cannot be moved to another group*.

10.9

Non-Hierarchical Agglomerative

- The number of groups is defined 'a priori'.
- Objects are assigned to groups until the criterion used to define the aggregation is optimized.
- The algorithm rule allows that a subject assigned to a group at a certain step can be moved to another group at a subsequent step.

10.10

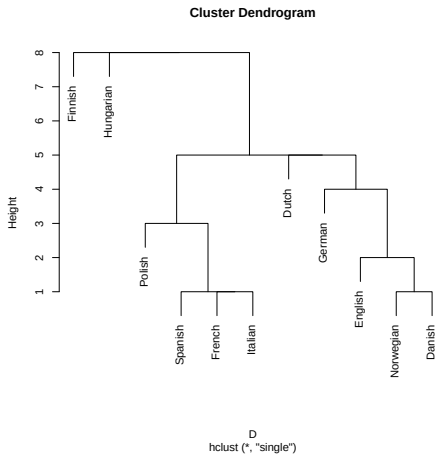
3.1 Hierarchical Methods

Let **D** be a $n \times n$ dissimilarity matrix, whose generic element d_{ij} is the dissimilarity between objects i and j.
D resumes dissimilarities among all n initial objects (clusters).

1. Search **D** for the nearest pair R, S of most similar clusters and merge R and S, by labeling the new cluster as $\{RS\}$.
2. Update the entries in the dissimilarity matrix **D** by (a) deleting rows and columns corresponding to clusters R and S and (b) adding a row and column giving distances between $\{RS\}$ and all the remaining objects calculated with *an appropriate criterion*.
3. Repeat $(n-1)$ times steps 1 and 2 until all n initial clusters are collapsed in a unique cluster.
   Record the identity of clusters that are merged and the levels (distances or similarities) at which the mergers take place.

The final aggregation scheme can be represented with a dendrogram.

10.11



10.12

A dendrogram is a graph that illustrates the mergers made at successive levels. It has the following properties.

- It is a tree chart, vertical or horizontal, that represents successive partitions.
- The *leaves* of the tree are the initial objects.
- The graph shows distances at which clusters (objects) are merged.
  The height of each branch in the plot is proportional to the value of the intergroup dissimilarity between its two parents/daughters (the leaves representing individual observations are all plotted at zero height).
- The dendrogram shows partitions that can be obtained at increasing distance levels.
- A 'cut' might be suggested where there is the longest branch (in order to find an optimum number of groups).
- More reasonable cut points might exist.
- The last 'leaves' that merge to the unique final cluster might be outlier objects.

10.13

Observe that there is no definitive answer on where to cut the dendrogram, since cluster analysis is essentially an exploratory approach; the interpretation of the resulting hierarchical structure is context-dependent and often several solutions are equally good from a theoretical point of view.

10.14

We have to define what does *an appropriate criterion* at step 2 of the hierarchical methods algorithm mean.

Usually one of the following criteria is used.

- Single linkage (Nearest-Neighbor)
- Complete linkage (Farthest-Neighbor)
- Average linkage
- Ward (incremental sum of squares)

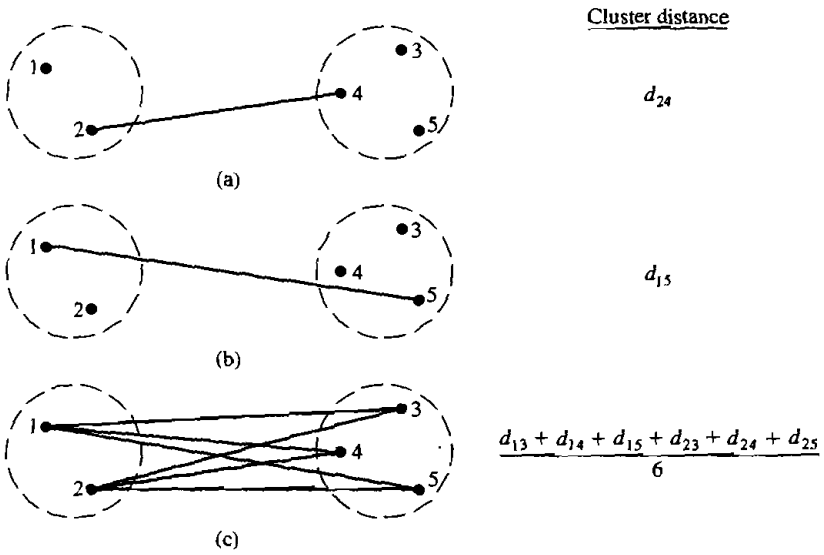


Figure 12.2 Intercluster distance (dissimilarity) for (a) single linkage, (b) complete linkage, and (c) average linkage.

Figure extracted from Johnson and Wichern 2007, Applied Multivariate Statistical Analysis.

10.15

Single linkage (Nearest-Neighbor)

The distance between a group $\{RS\}$ and an object (or group) Q is defined as

$$d_{\{RS\},Q} = \min(d_{RQ}, d_{SQ}).$$

10.16

For example let

$$\mathbf{D} = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & & & \\ b & 1 & 0 & & \\ c & 3 & 2 & 0 & \\ d & 7 & 6 & 4 & 0 \end{array}$$

The first group will be $\{ab\}$
The criterion $d_{\{RS\},Q} = \min(d_{RQ}, d_{SQ})$ updates matrix $\mathbf{D}$ as

$$\mathbf{D}_1 = \begin{array}{c|ccc} & ab & c & d \\ \hline ab & 0 & & \\ c & 2 & 0 & \\ d & 6 & 4 & 0 \end{array}$$

The second group will be $\{abc\}$
The single linkage criterion updates $\mathbf{D}_1$ as

$$\mathbf{D}_2 = \begin{array}{c|cc} & abc & d \\ \hline abc & 0 & \\ d & 4 & 0 \end{array}$$

The final group is $\{abcd\}$. 10.17

Complete linkage (Farthest-Neighbor)

The distance between a a group $\{RS\}$ and an object (or group) Q is defined as

$$d_{\{RS\},Q} = \max(d_{RQ}, d_{SQ}).$$

10.18

For example let

$$\mathbf{D} = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & & & \\ b & 1 & 0 & & \\ c & 3 & 2 & 0 & \\ d & 7 & 6 & 4 & 0 \end{array}$$

The first group will be $\{ab\}$
The criterion $d_{\{RS\},Q} = \max(d_{RQ}, d_{SQ})$ updates matrix $\mathbf{D}$ as

$$\mathbf{D}_1 = \begin{array}{c|ccc} & ab & c & d \\ \hline ab & 0 & & \\ c & 3 & 0 & \\ d & 7 & 4 & 0 \end{array}$$

The second group will be $\{abc\}$
The complete linkage criterion updates $\mathbf{D}_1$ as

$$\mathbf{D}_2 = \begin{array}{c|cc} & abc & d \\ \hline abc & 0 & \\ d & 7 & 0 \end{array}$$

The final group is $\{abcd\}$. 10.19

Average linkage

Average linkage treats the distance between two clusters as the average distance between all pairs of items where one member of a pair belongs to each cluster.
Therefore,the distance between a cluster $\{RS\}$ and an object (or group) Q is defined as

$$d_{\{RS\},Q} = \frac{\sum_i \sum_k d_{ik}}{n_{\{RS\}} n_Q}$$

where d_{ik} is the distance between object i in cluster $\{RS\}$ and object k in the cluster Q, and $n_{\{RS\}}$ and n_Q are the number of items in clusters $\{RS\}$ and Q, respectively. 10.20
For example let

$$\mathbf{D} = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & & & \\ b & 1 & 0 & & \\ c & 3 & 2 & 0 & \\ d & 7 & 6 & 4 & 0 \end{array}$$

The first cluster will be $\{ab\}$
The criterion $d_{\{RS\},Q} = \frac{\sum_i \sum_k d_{ik}}{n_{\{RS\}} n_Q}$ updates matrix $\mathbf{D}$ as

$$\mathbf{D}_1 = \begin{array}{c|ccc} & ab & c & d \\ \hline ab & 0 & & \\ c & (3+2)/2 & 0 & \\ d & (7+6)/2 & 4 & 0 \end{array}$$

The second cluster will be $\{abc\}$
The average linkage criterion updates $\mathbf{D}_1$ as

$$\mathbf{D}_2 = \begin{array}{c|cc} & abc & d \\ \hline abc & 0 & \\ d & (7+6+4)/3 & 0 \end{array}$$

The final cluster is $\{abcd\}$. 10.21

Ward (incremental sum of squares)

Ward considered hierarhical clustering procedures based on minimizing the 'loss of information' from joining two groups.
The method is usually implemented with loss of information taken to be as an increase in an error sum of squares ESS. 10.22
For a given cluster k, let ESS_k be the sum of the squared deviations of every object in the cluster from the cluster mean (centroid).
If there are K cluster let $ESS = \sum_{k=1}^{K} ESS_k$.

At each step in the analysis, the union of every possible pair of clusters is considered, and the two clusters whose combination results in the smallest increase in ESS (minimum loss of information) are joined.

ESS ranges between $\min ESS = 0$ when all clusters consist of single objects and $\max ESS = \sum_{i=1}^{n} (y_i - \bar{y})'(y_i - \bar{y}))$ when all objects are combined in a single group. 10.23

Which criterion?

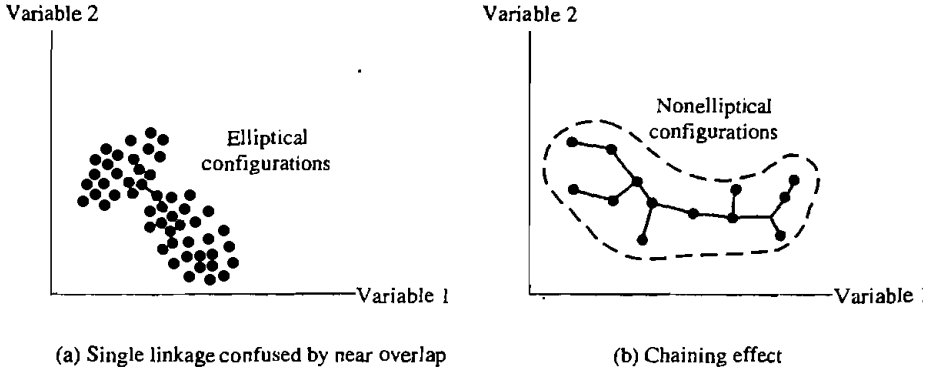
There is no unique answer, but the choice can be made according to the so-called minimal well-structured partition.

Minimal well-structured Partition
An agglomerative method should have some properties:

- The maximum distance within clusters has to be lower than the minimum distance between clusters, and the number of clusters that ensures this property must be as small as possible.
- The partition must be invariant to monotone trasformations of distances (this holds for the single and the complete linkages)

10.24

Complete linkage is commonly used. Its drawback is its ability in finding elliptical groups. The single linkage can finds also non elliptical structures but can be confused in presence of near overlapping of elliptical structures, and has the so-called chaining effect drawback; it can merge very far objects if these are joined by a sequence of intermediate objects.



(a) Single linkage confused by near overlap (b) Chaining effect

Figure 12.5 Single linkage clusters.

Figure extracted from Johnson and Wichern 2007, Applied Multivariate Statistical Analysis.

10.25

3.2 Non-Hierarchical Methods (K-means)

In its simplest version, the classification process is composed of these three steps.

1. K initial points, named centroids, are first selected. K values in $\Re^p$, the p-dimensional euclidean space.
2. Proceed through the list of items, assigning an item to the cluster whose centroid (mean) is nearest. (Distance is usually computed using Euclidean distance with either standardized or unstandardized observations.)
   Recalculate the centroid for the cluster receiving the new item and for the cluster losing the item.
3. Repeat step 2 until no more reassignments take place.

10.26

For example let consider the following data

	X_1	X_2
a	5	3
b	-1	1
c	1	-2
d	-3	-2

Let the initial centroid be given by the averages of groups $\{ab\}$ and $\{cd\}$ (this assumes that we have already decided that the final solution is made up by 2 groups); the initial solution is characterized by centroids

	centroids	
cluster	$\bar{x}_1$	$\bar{x}_2$
ab	2	2
cd	-1	-2

10.27

Let us compute the square Euclidean distances of each initial object (a, b, c and d) from the group centroids. We have

$$d^2_{\{ab\},a} = (5-2)^2 + (3-2)^2 = 10$$
$$d^2_{\{cd\},a} = (5+1)^2 + (3+2)^2 = 61$$

Has a to be reassigned?
Since a is closer to $\{ab\}$ than to $\{cd\}$ it is not reassigned.

$$d^2_{\{ab\},b} = (-1-2)^2 + (1-2)^2 = 10$$
$$d^2_{\{cd\},b} = (-1+1)^2 + (1+2)^2 = 9$$

Has b to be reassigned?
Yes, b is reassigned to cluster $\{cd\}$, giving cluster $\{bcd\}$ and the following updated coordinates of the centroids

	centroids	
cluster	$\bar{x}_1$	$\bar{x}_2$
a	5	3
bcd	-1	-1

10.28

Again each item is checked for reassignment. Computing the squared Euclidean distances gives the following

	objects			
cluster	a	b	c	d
a	0	40	41	89
bcd	52	4	5	5

No object needs to be reassigned and the algorithm stops.

To check the stability of the clustering, it is desirable to rerun the algorithm with a new initial partition.

10.29

Advantages and disadvantages

- It can be used to group very big data sets ($n \simeq 1,000,000$).
- It does not show how objects are successively aggregated.
- It is a fast method (has low computation times).
- It can be repeated with different number of groups.
- Its use is suggested when one is more interested in creating *types of objects* than in aggregating single objects.

In presence of big data sets guesses of the number of groups and initial centroids can be obtained by applying hierarchical cluster analysis to a small sample of observations representative of all data.
Namely, there are strong arguments for not fixing the number of clusters, K, in advance, including the following:

10.30

- If two or more seed points (initial clusters) inadvertently lie within a single cluster, their resulting clusters will be poorly differentiated.
- The existence of an outlier might produce at least one group with very disperse items.
- Even if the population is known to consist of K groups, the sampling method may be such that data from the rarest group do not appear in the sample. Forcing the data into K groups would lead to nonsensical clusters.

10.31

4 Examples

We apply clustering to the following problems

- classification of US public utilities (both variables and cases)
- classification of languages
- classification of Scotch whiskies

10.32

4.1 Classification of US Public Utilities (Both Variables and Cases)

The data set `publiccompanies.dat` contains information about 22 US public utilities

- X_1: fixed-charge coverage ratio (income/debt)
- X_2: rate of return on capital
- X_3: cost of KW capacity in place
- X_4: annual load factor
- X_5: peak kWh demand growth from 1974 to 1975
- X_6: sales (kWh use per year)
- X_7: percent nuclear
- X_8: total fuel costs (cents per kWh)

10.33

```
pubcomp <- read.table("publiccompanies.dat")
head(pubcomp)
##     V1   V2  V3   V4   V5    V6   V7    V8       V9
## 1 1.06  9.2 151 54.4  1.6  9077  0.0 0.628  Arizona
## 2 0.89 10.3 202 57.9  2.2  5088 25.3 1.555   Boston
## 3 1.43 15.4 113 53.0  3.4  9212  0.0 1.058  Central
## 4 1.02 11.2 168 56.0  0.3  6423 34.3 0.700   Common
## 5 1.49  8.8 192 51.2  1.0  3300 15.6 2.044 Consolid
## 6 1.32 13.5 111 60.0 -2.2 11127 22.5 1.241  Florida
rownames(pubcomp) <- pubcomp[, 9]
pubcomp <- pubcomp[, -9]
```

10.34

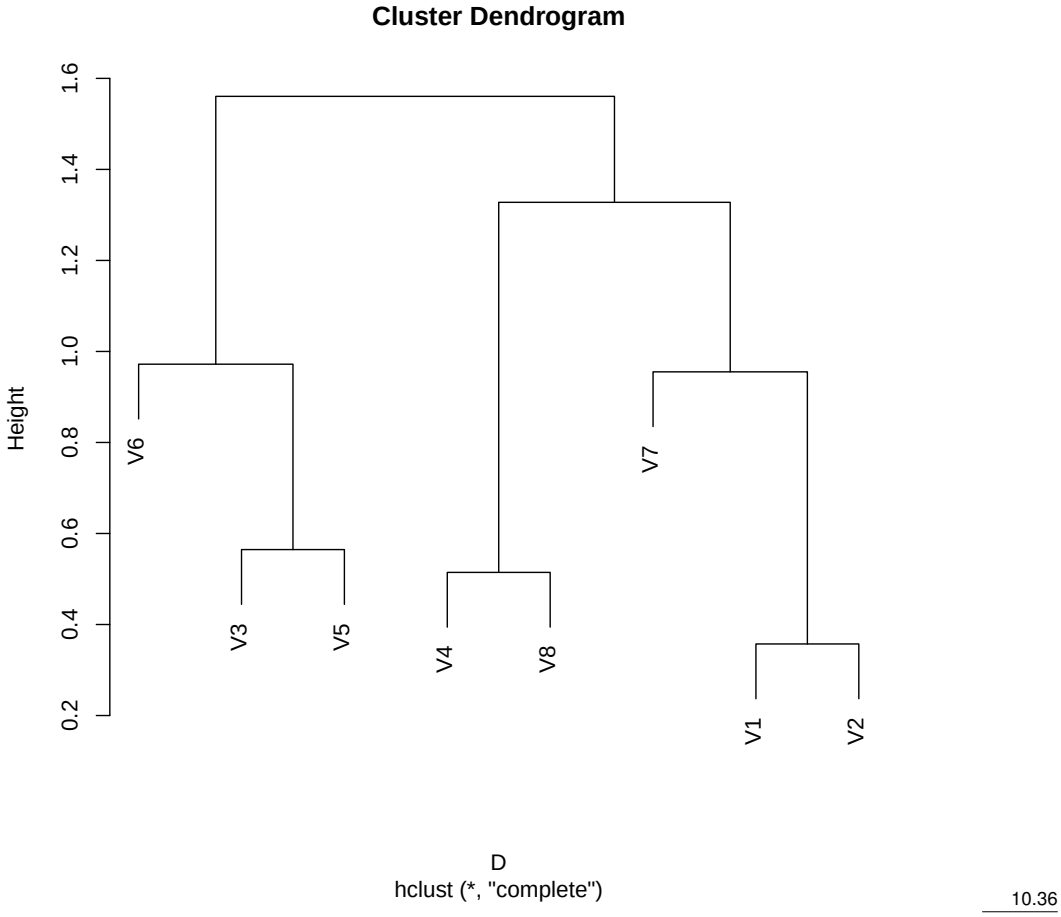
Example of clustering of variables

```
round(pubcomp.cor <- cor(pubcomp), 3)
##        V1     V2     V3     V4     V5     V6     V7     V8
## V1  1.000  0.643 -0.103 -0.082 -0.259 -0.152  0.045 -0.013
## V2  0.643  1.000 -0.348 -0.086 -0.260 -0.010  0.211 -0.328
## V3 -0.103 -0.348  1.000  0.100  0.435  0.028  0.115  0.005
## V4 -0.082 -0.086  0.100  1.000  0.033 -0.288 -0.164  0.486
## V5 -0.259 -0.260  0.435  0.033  1.000  0.176 -0.019 -0.007
## V6 -0.152 -0.010  0.028 -0.288  0.176  1.000 -0.374 -0.561
## V7  0.045  0.211  0.115 -0.164 -0.019 -0.374  1.000 -0.185
## V8 -0.013 -0.328  0.005  0.486 -0.007 -0.561 -0.185  1.000
```

```
D <- 1 - as.dist(pubcomp.cor)
pubcomp.cor.complete <- hclust(D, method = "complete")
```

10.35

```
plot(pubcomp.cor.complete)
```

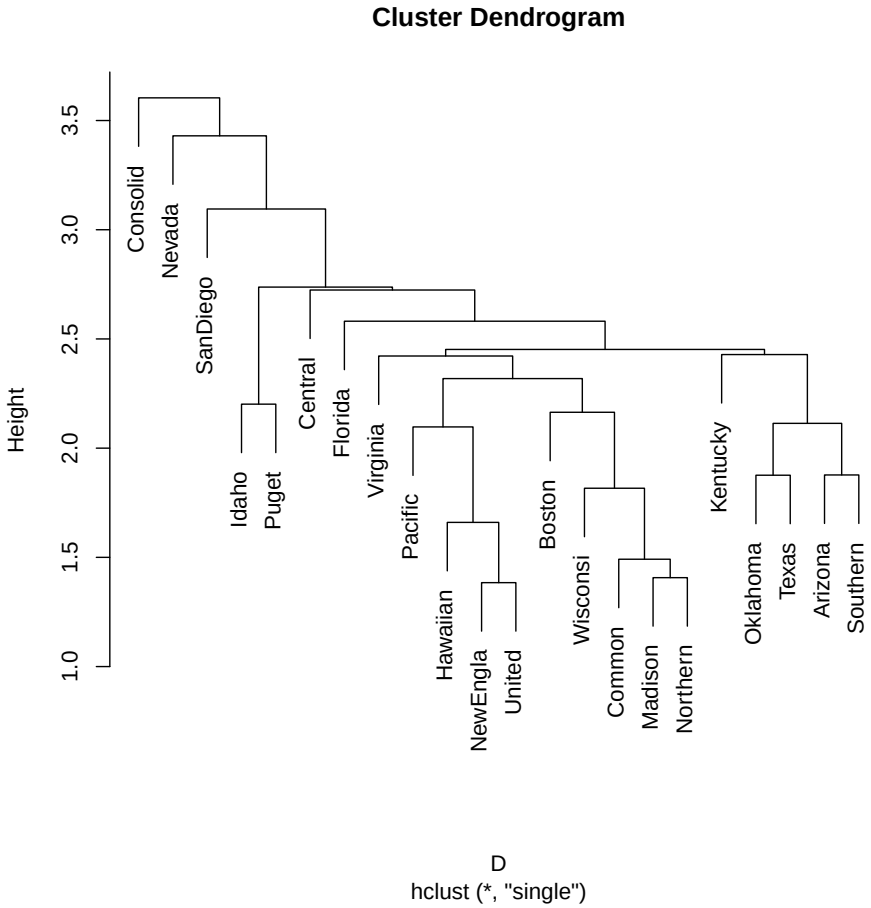


Example of clustering of cases (we consider standardized variables)

```
D <- dist(scale(pubcomp))
```

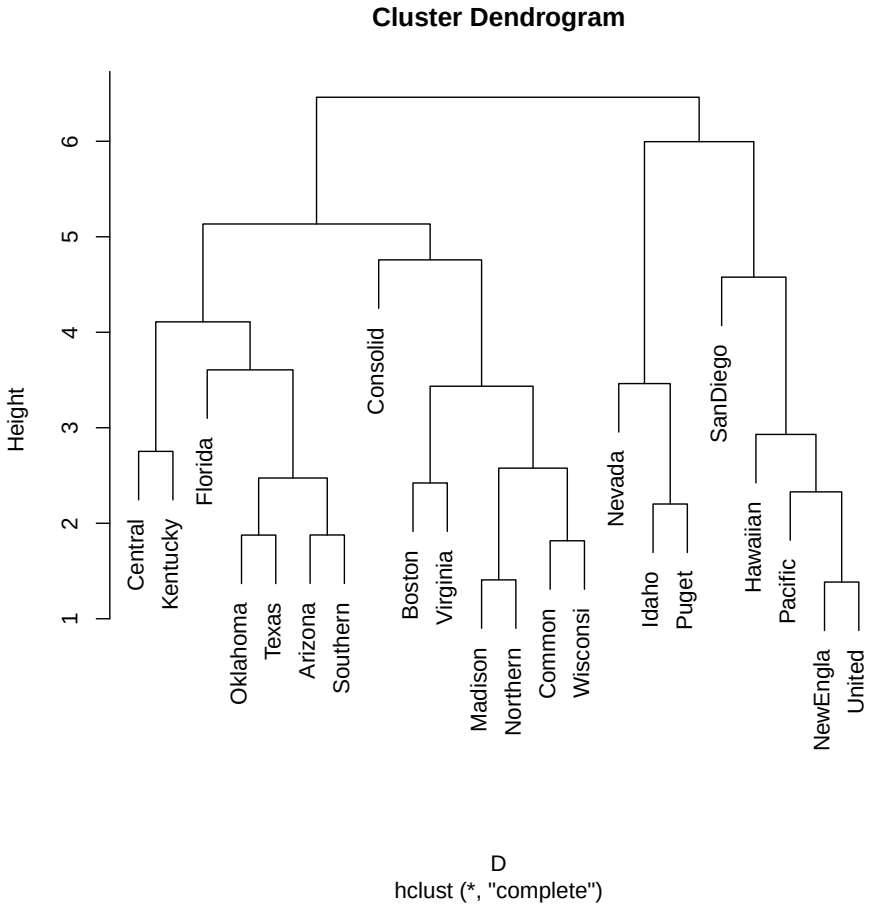
10.37

```
pubcomp.single <- hclust(D, method = "single")
plot(pubcomp.single)
```



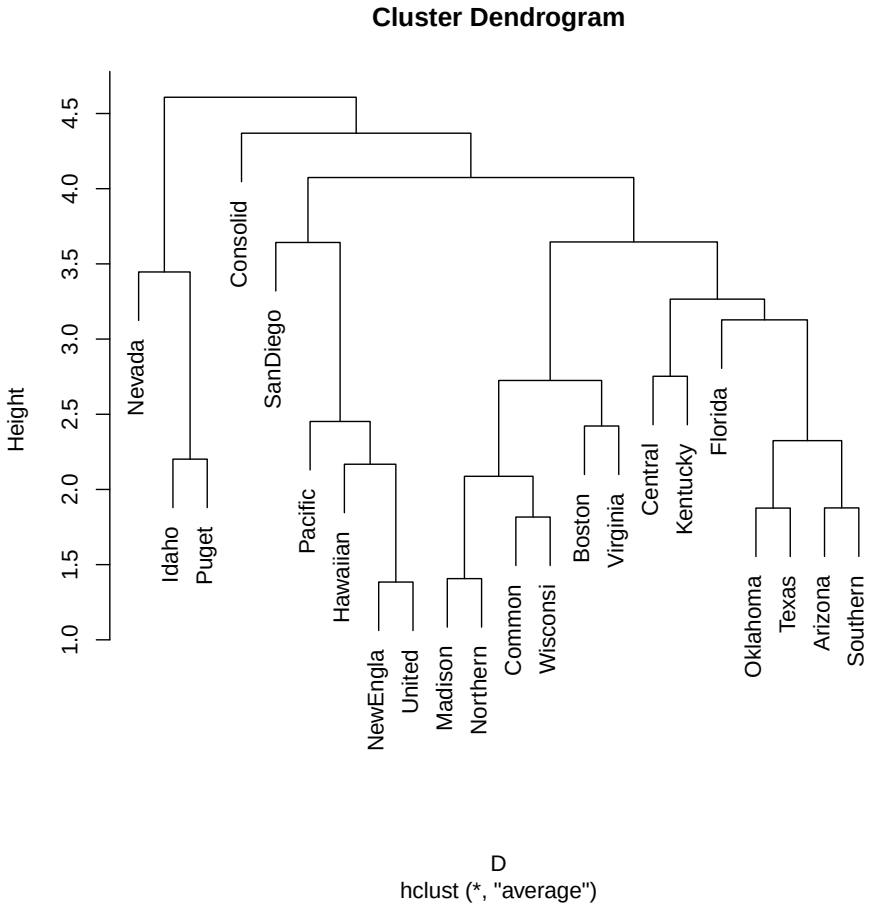
10.38

```
pubcomp.complete <- hclust(D, method = "complete")
plot(pubcomp.complete)
```



10.39

```r
pubcomp.average <- hclust(D, method = "average")
plot(pubcomp.average)
```



10.40

4.2 Classification of Languages

A very simple example of textual considers the classification of languages according to their starting letters of the first 10 numerals.

```
lang <- read.table("numerals.txt", header = TRUE, sep = ",",
    stringsAsFactors = FALSE)
```

10.41

```
lang
##    English Norwegian Danish Dutch German French Spanish
## 1      one        en     en   een    ein     un     uno
## 2      two        to     to  twee   zwei   deux     dos
## 3    three       tre    tre  drie   drei  trois    tres
## 4     four      fire   fire  vier   vier quatre  cuatro
## 5     five      fern    fem  vijf   funf   cinq   cinco
## 6      six      seks   seks   zes  sechs    six    seix
## 7    seven       sju    syv zeven sieben   sept   siete
## 8    eight      atte   otte  acht   acht   huit    ocho
## 9     nine        ni     ni negen   neun   neuf   nueve
## 10     ten        ti     ti  tien   zehn    dix    diez
##    Italian   Polish Hungarian   Finnish
## 1      uno    jeden       egy      yksi
## 2      due      dwa     ketto     kaksi
## 3      tre     trzy     harom     kolme
## 4  quattro   cztery      negy      neua
## 5   cinque     piec        ot     viisi
## 6      sei    szesc       hat     kuusi
## 7    sette   siedem       het  seiseman
## 8     otto    osiem     nyolc kahdeksan
## 9     nove dziewiec    kilenc  ybdeksan
## 10   dieci dziesiec       tiz  kymmenen
```

10.42

```
lang <- t(as.matrix(lang))
nlang <- nrow(lang)
initials <- substr(as.matrix(lang), 1, 1)
init.conc <- lapply(1:nlang, {
    function(j) sapply(1:nlang, {
        function(i) sum(initials[j, ] == initials[i, ])
    })
})
concordances <- matrix(unlist(init.conc), nlang, nlang)
rownames(concordances) <- colnames(concordances) <- rownames(initials)
```

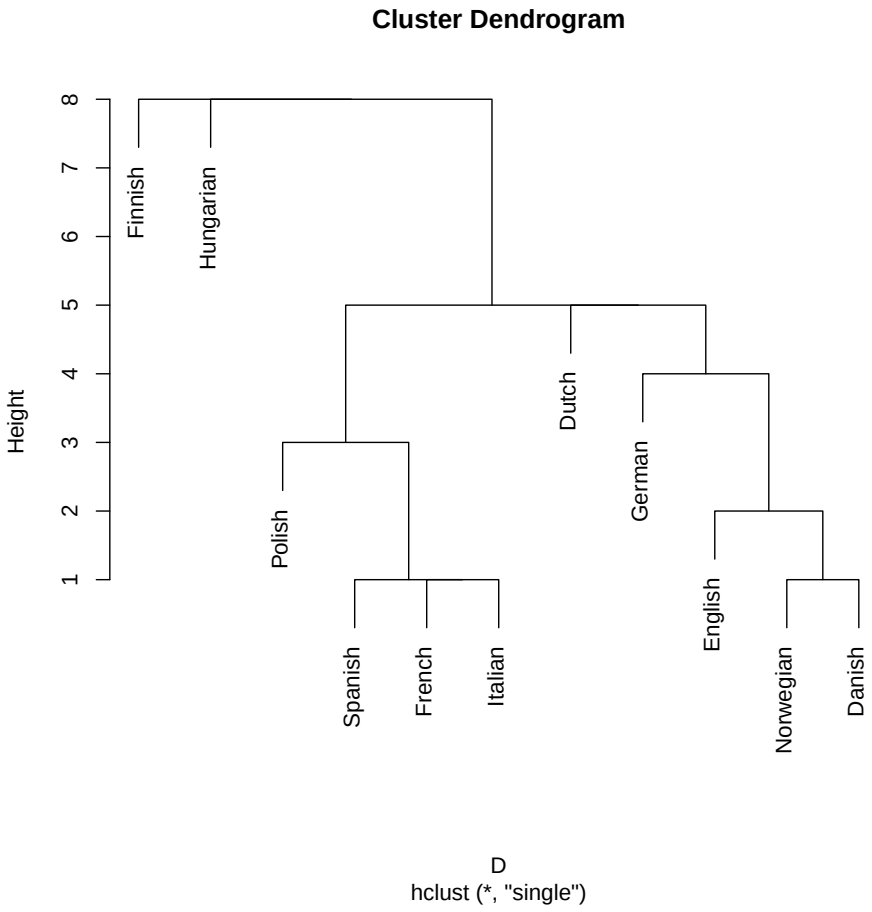
10.43

```
10 - concordances
##           English Norwegian Danish Dutch German French
## English         0         2      2     7      6      6
## Norwegian       2         0      1     5      4      6
## Danish          2         1      0     6      5      6
## Dutch           7         5      6     0      5      9
## German          6         4      5     5      0      7
## French          6         6      6     9      7      0
## Spanish         6         6      5     9      7      2
## Italian         6         6      5     9      7      1
## Polish          7         7      6    10      8      5
```

```
## Hungarian       9         8      8     8      9     10
## Finnish         9         9      9     9      9      9
##           Spanish Italian Polish Hungarian Finnish
## English         6       6      7         9       9
## Norwegian       6       6      7         8       9
## Danish          5       5      6         8       9
## Dutch           9       9     10         8       9
## German          7       7      8         9       9
## French          2       1      5        10       9
## Spanish         0       1      3        10       9
## Italian         1       0      4        10       9
## Polish          3       4      0        10       9
## Hungarian      10      10     10         0       8
## Finnish         9       9      9         8       0
D <- as.dist(10 - concordances)
D
##           English Norwegian Danish Dutch German French
## Norwegian       2
## Danish          2         1
## Dutch           7         5      6
## German          6         4      5     5
## French          6         6      6     9      7
## Spanish         6         6      5     9      7      2
## Italian         6         6      5     9      7      1
## Polish          7         7      6    10      8      5
## Hungarian       9         8      8     8      9     10
## Finnish         9         9      9     9      9      9
##           Spanish Italian Polish Hungarian
## Norwegian
## Danish
## Dutch
## German
## French
## Spanish
## Italian         1
## Polish          3       4
## Hungarian      10      10     10
## Finnish         9       9      9         8
```

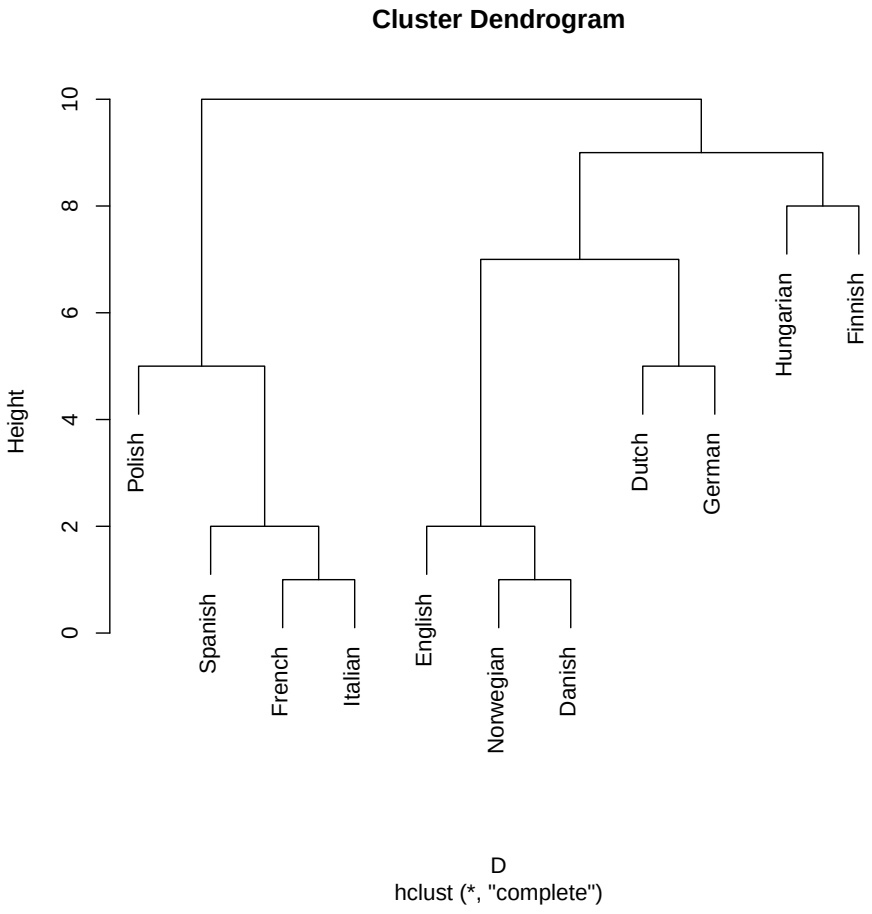
10.44

```r
lang.single <- hclust(D, method = "single")
plot(lang.single)
```



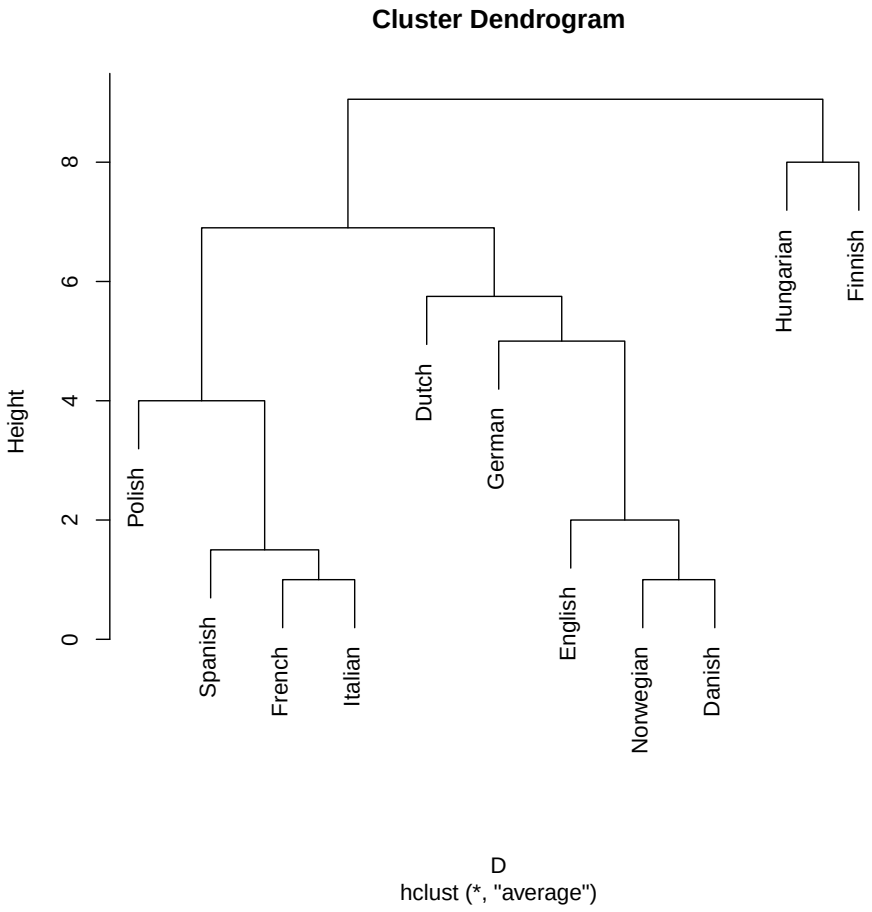
10.45

```
lang.complete <- hclust(D, method = "complete")
plot(lang.complete)
```



10.46

```
lang.average <- hclust(D, method = "average")
plot(lang.average)
```



10.47

4.3 Classification of Scotch Whiskies

LaPointe and Legendre (1994) A classification of Pure Malt Scotch Whiskies *Applied Statistics*, 43, 237-257 show motivation and results of the present example.

Geographical reference of results is reported only to show how to build a map. A clustering with spatial contiguity constraint has to be performed to correctly analyze geographical information.

10.48

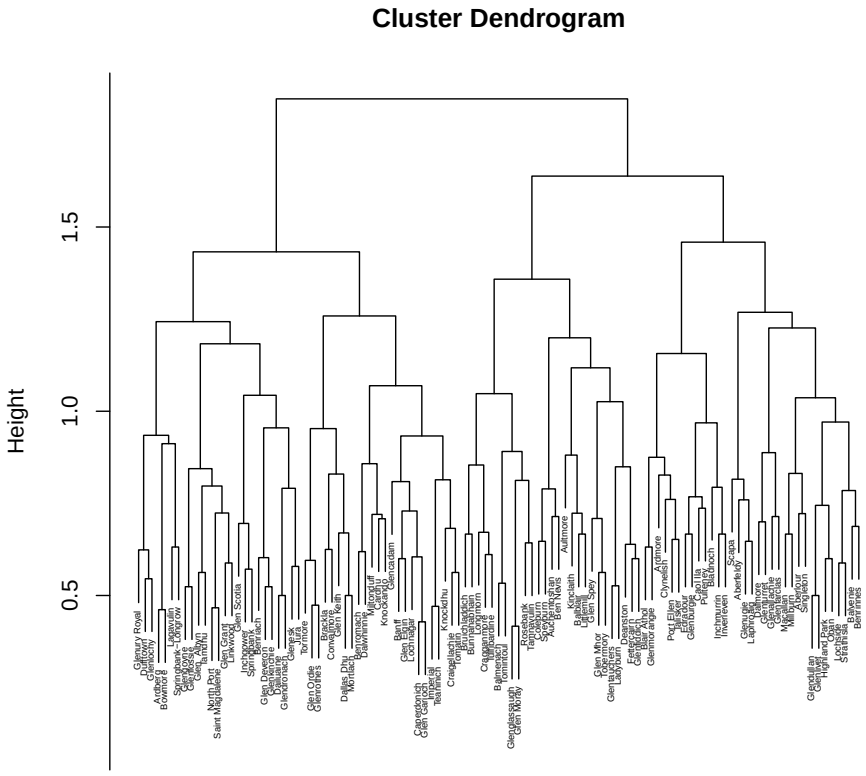
```
data <- read.table("SCOTCH.TXT")
colnames(data) <- c(paste("color", 1:14), paste("nose", 1:12),
    paste("body", 1:8), paste("pal", 1:15), paste("fin", 1:19))
distillery <- read.table("DISTCOOR.TXT", sep = "\t", row.names = 1)
colnames(distillery) <- c("longitude", "latitude")
rownames(data) <- rownames(distillery)

D <- proxy::dist(data, "jaccard")

ave.clus <- hclust(D, method = "ward.D2")
```

10.49

```r
plot(ave.clus, cex = 0.4)
```



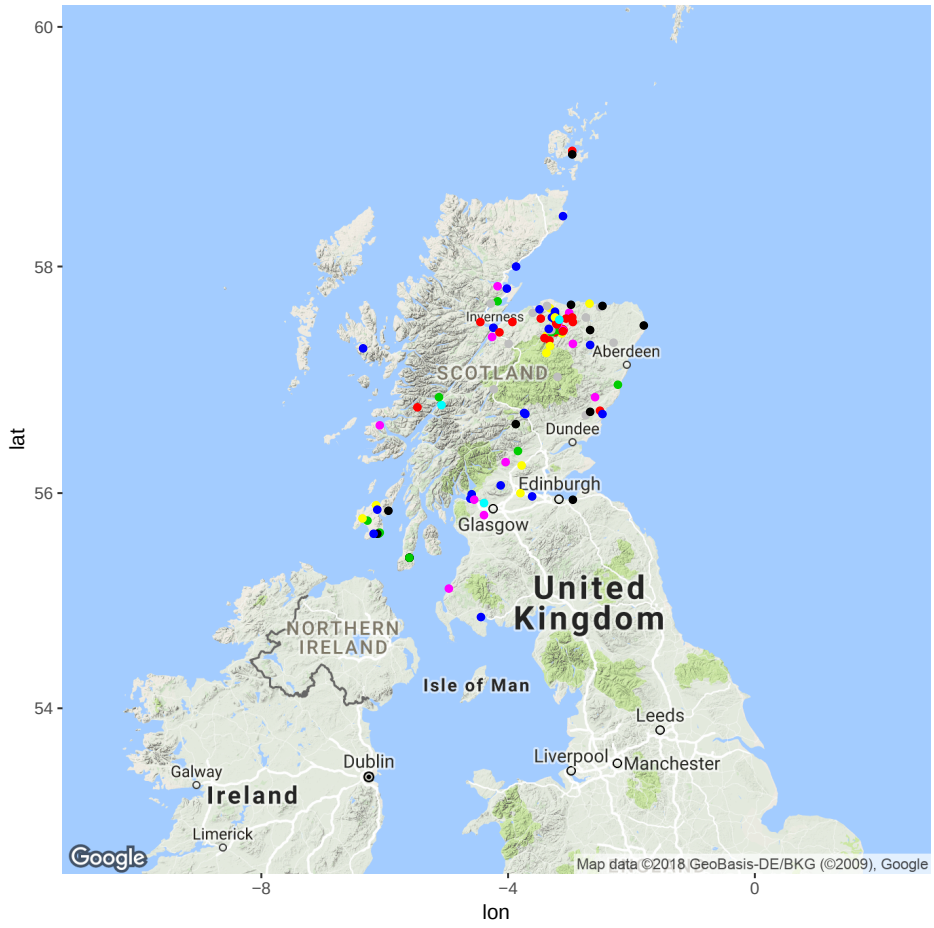

```r
group <- cutree(ave.clus, k = 12)
table(group)
```

```
## group
##  1  2  3  4  5  6  7  8  9 10 11 12
##  4 12  7 13  4 12 11 16 10  8  4  8
```

The package `ggmap` allows to download maps from Googlemaps[1].

```
library(ggmap)
map <- get_map(location = "scotland", zoom = 6)
mapPoints <- ggmap(map) + geom_point(aes(x = -longitude, y = latitude),
    data = distillery, color = group)
mapPoints
```

10.52



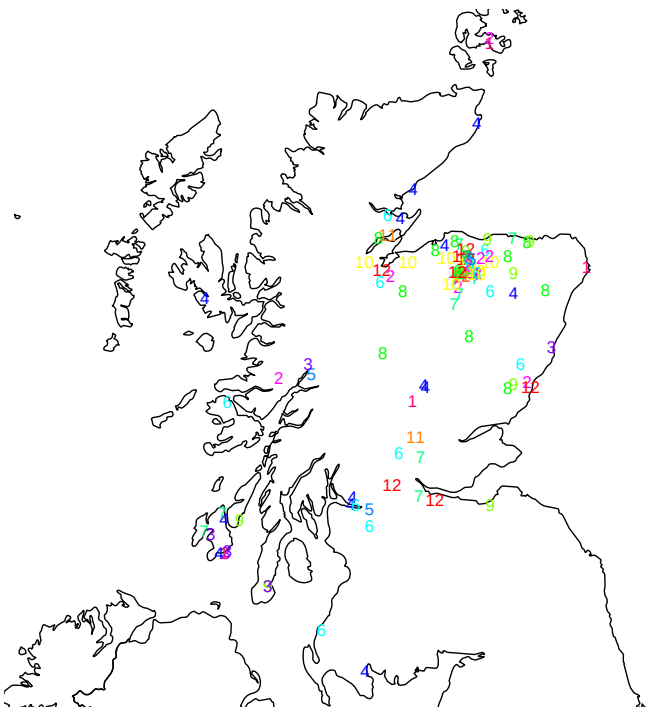
10.53

[1] As remarked by David Kahle 'As of mid-2018, the Google Maps Platform requires a registered API key. While this alleviates previous burdens (e.g. query limits), it creates some challenges as well. The most immediate challenge for most R users is that ggmap functions that use Google's services no longer function out of the box, since the user has to setup an account with Google, enable the relevant APIs, and then tell R about the user's setup.' Please, read carefully the Details section of the `?ggmap::register_google` help page, before proceeding with setting up a Google account.

If you do not have an account for Google cloud services, the package `rworldmap` contains maps for the whole world.

```
library(rworldmap)
newmap <- getMap(resolution = "high")
plot(newmap, xlim = c(-6.4, -1.8), ylim = c(54.8, 59))
with(distillery, {
    text(-longitude, latitude, group, col = rev(rainbow(12))[group],
        cex = 0.75)
})
```

10.54



10.55